

The University of the South Pacific

School of Computing, Information & Mathematical Sciences

CS 112 Data Structures and Algorithms

Tutorial Week 5

In this tutorial you are required to use class to make your previous tutorial program modular and more manageable. After completion of this tutorial see how classes simplify the program by giving a proper structure to your program.

Question 1. (Allocated Time: 90 mins)

Week 2 tutorial has been modified (simplified) to be used in this tutorial. Download, read and run modified week 2 tutorial from *week5: Object Oriented Concepts / Tutorial* folder.

Do the following changes:

- i. Create a class for student so that it can encapsulate marks, id and grade. Make sure you have constructor, destructor and all setter and getter methods.
- ii. Delete all parallel arrays.
- iii. Instead of parallel arrays use just one array of student class for this program. Do all the required changes.

Modify function *void print_students_records* to pass an object of student instead of passing students' information separately. Syntax for the prototype of modified function should be *void print_students_record (student s[], int size)*.

Do you see now you don't have several parallel arrays floating around in your program?

Question 2. (Allocated Time: 20 mins)

Download instructions from *week5: Object Oriented Concepts / Tutorial* folder to create a project in Dev C++ for the program you made in question 1.

Question 3. (Home work question)

Delete *char determine_grade(int marks)* function and make it a behavior of student class. Your prototype should become **void** *determine_grade()*. Better name for this method could also be *void assign_grade()* or *void allocate_grade()*.